Multi-Tenant RAG Chatbot Platform

Complete Documentation Suite

1. User Guide

1.1 Introduction

The Multi-Tenant RAG Chatbot System is an enterprise-grade SaaS platform designed to let multiple business tenants construct, train, customize, and deploy AI support assistants on their websites. Each business operates in complete database and vector store isolation, ensuring data privacy.

The primary purpose is to simplify customer support, lead capture, and website-specific information retrieval. By crawling a business's public website or importing custom Q&A items, the system builds a semantic knowledge base. A customized chat widget embedded on their pages uses this knowledge to respond to visitor inquiries instantly. If the AI cannot resolve the query, the chat seamlessly routes to human support representatives.

1.2 Onboarding & Registration

New tenants must register through the system registration page to provision isolated environments. Registration requires entering a name, email address, username, and a password of at least six characters. Upon registration, the system dynamically provisions a dedicated MongoDB database, a custom folder for vector embeddings, and generates API endpoints.

1.3 User Login

Once registered, business owners can log in using their username or email address and password. Upon successful validation, the backend issues a JSON Web Token (JWT) that manages user sessions and secures dashboard API calls.

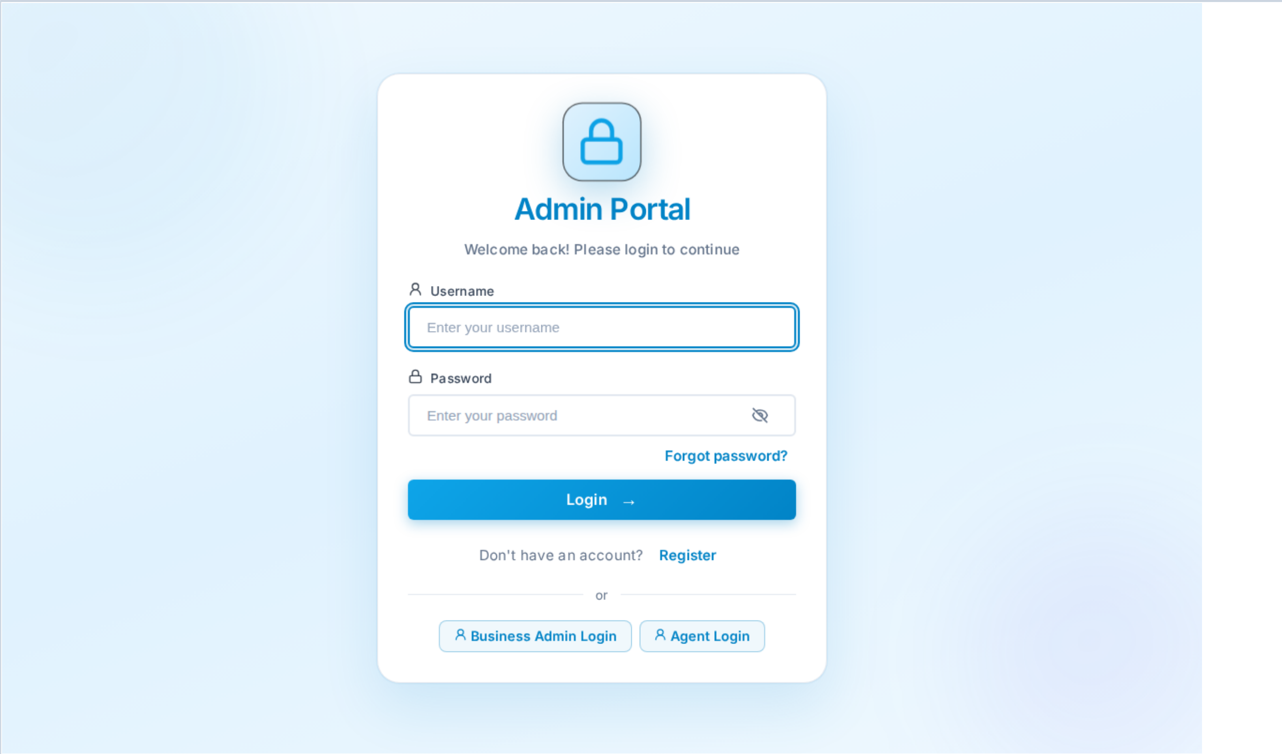

Figure 1.1: Platform Secure Login Portal

1.4 Dashboard Overview

The User Dashboard page acts as the central control hub. It lists total websites created, total conversations logged, leads captured, and active crawler schedulers. Business owners can review their websites, check scraping status, configure options, customize chat widget colors, or view recent leads in real-time.

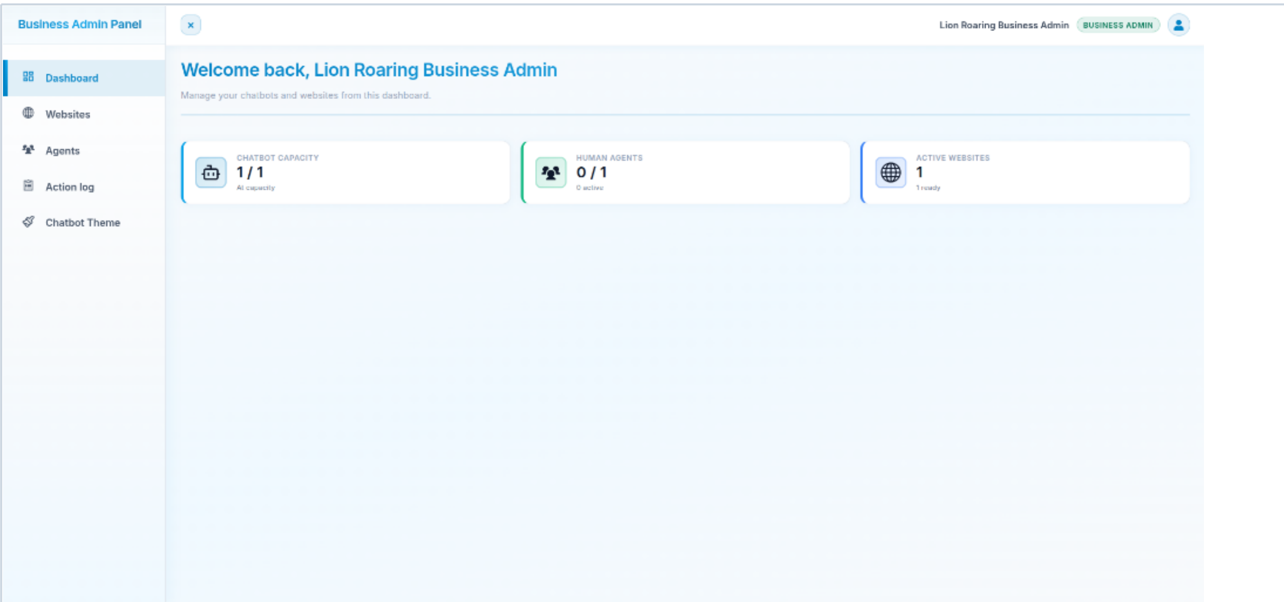

Figure 1.2: Business Admin Panel Dashboard

1.5 Website Management & Scraping

Tenants train the chatbot by indexing website URLs. Adding a website requires configuring a start URL and optional sitemap URL. Tenants can adjust settings such as depth limits (how deep the crawler follows internal links) and limit the maximum links crawled per page. For sites built on modern JavaScript frameworks, the Playwright rendering option compiles the page layout before extracting text. Scrapes can be triggered immediately or scheduled to run every two hours to keep the bot updated.

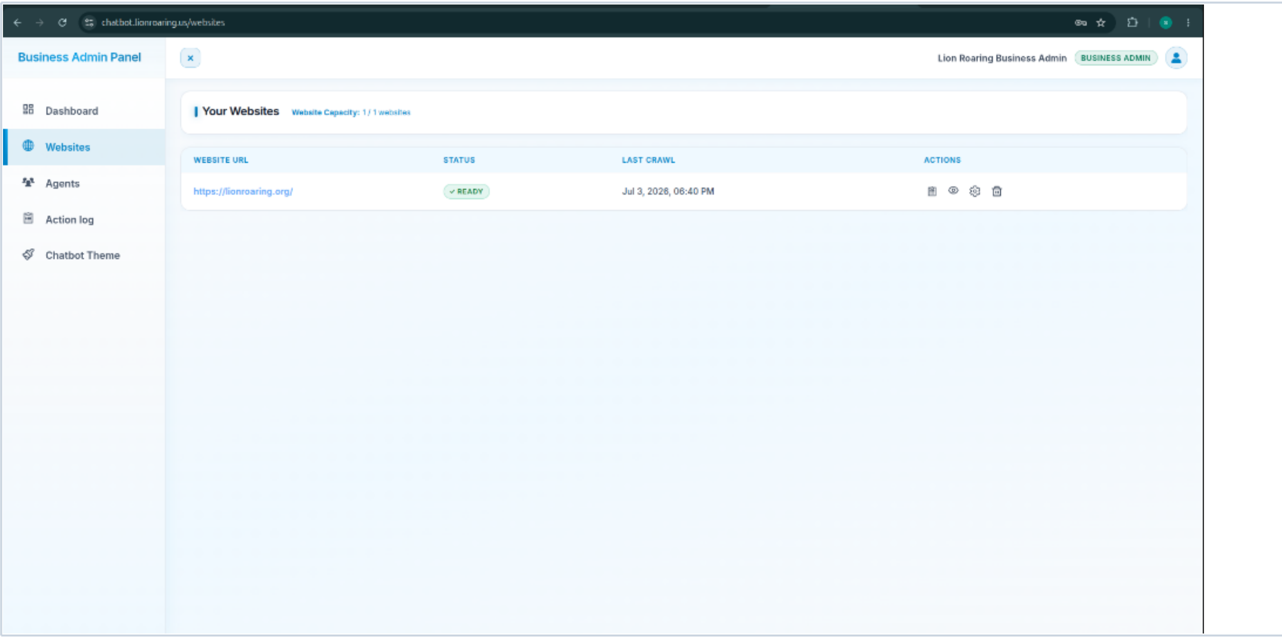

Figure 1.3: Websites Indexing Console

1.6 Crawling & Installation Workflow

From the Websites list, click the gear actions icon to open the Website Actions page. Here, you have three primary workflows:

- **Run Crawl:** Click this button to initiate website crawling. The crawler parses the website pages and adds the extracted content to the chatbot's knowledge base.
- **Add Knowledge:** Click this button to navigate to the Knowledge Base page, where you can manually add custom Q&A items to train your chatbot.
- **Install Widget:** Click this button to open the installation script pop-up and retrieve the script snippet for both web and mobile deployment.

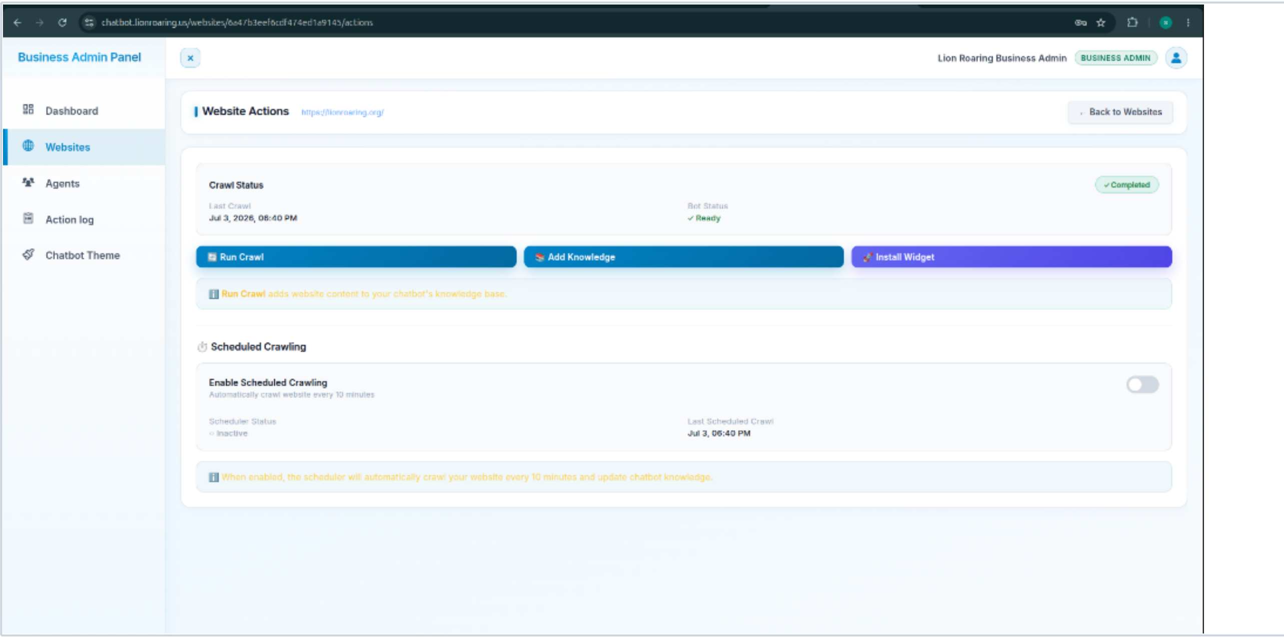

Figure 1.4: Website Actions Portal (Run Crawl, Add Knowledge, Install Widget)

1.6.1 Installing the Chatbot Widget

Clicking the "Install Widget" button displays a modal containing the widget configuration and an HTML script snippet. Copy this snippet and paste it into your website's HTML, just before the closing body tag. This snippet loads the widget, passes your bot ID and authentication token, and integrates the chatbot on your site.

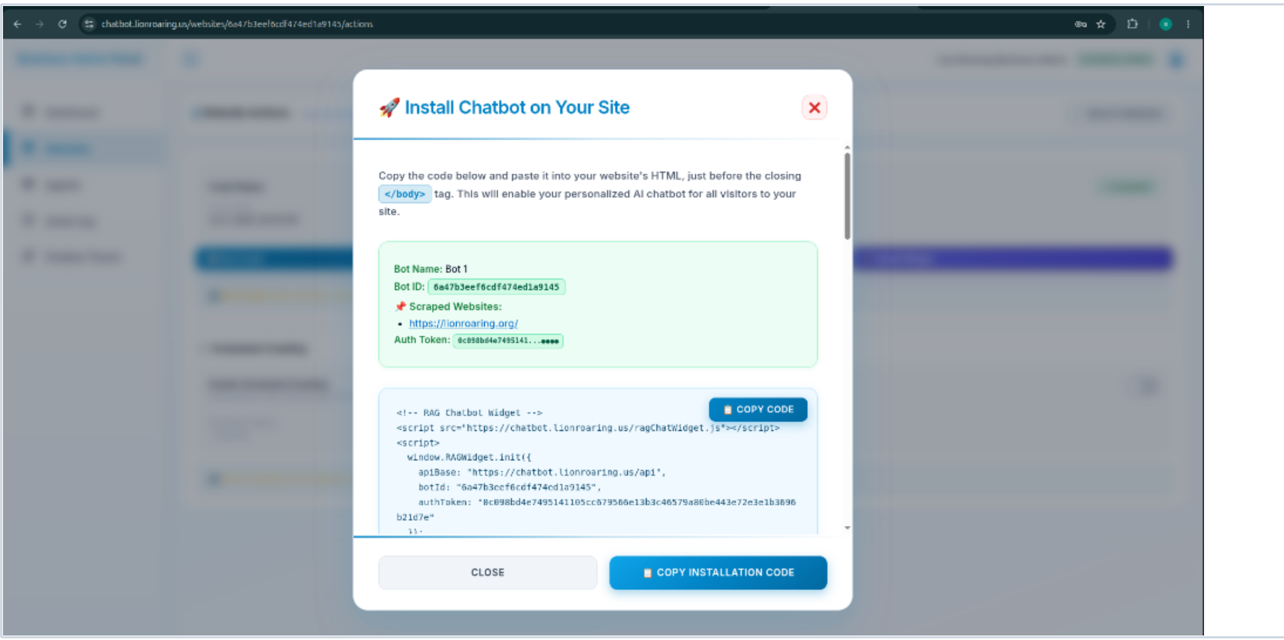

Figure 1.5: Web and Mobile Installation Script Modal

1.6.2 Adding Manual Knowledge

If certain information is not available on your website or if you want to override default answers, click "Add Knowledge". This opens the Knowledge Base page where you can manually input custom Q&A pairs (questions and answers) directly. These items are indexed instantly, training the chatbot's expertise alongside scraped website pages.

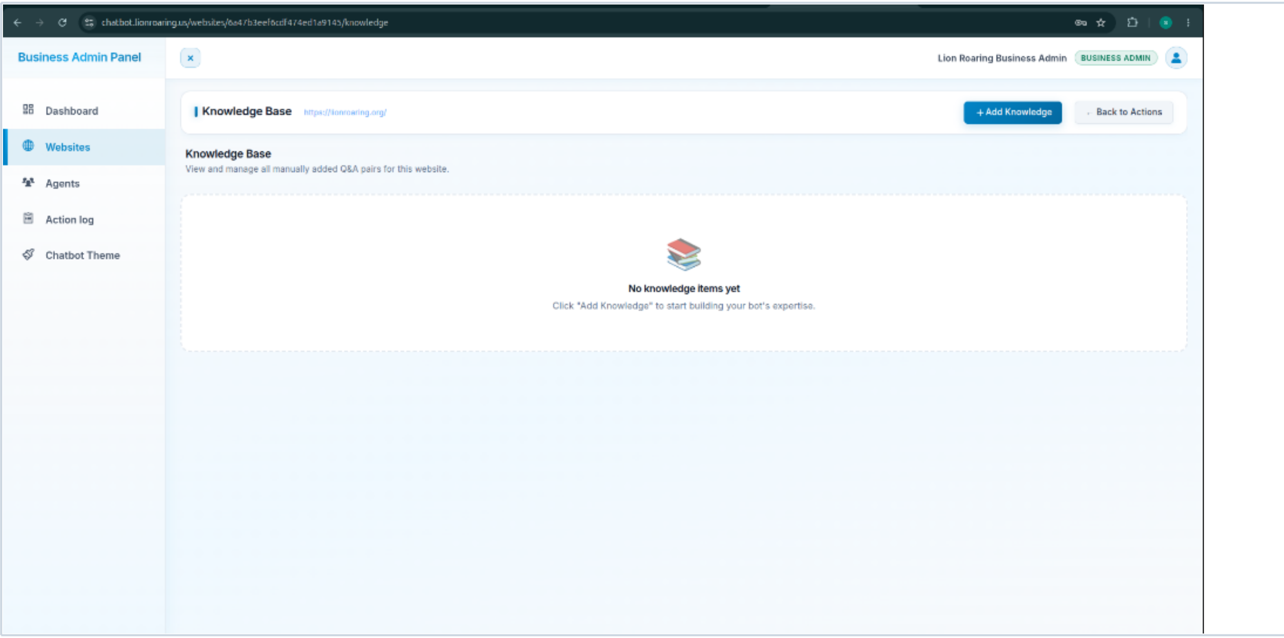

Figure 1.6: Manual Knowledge Base Editor

> **Important Limitation**
>
> The crawler ignores document downloads exceeding 5 megabytes. It cannot bypass verification gates like CAPTCHAs or pages protected by login screens.

1.7 Live Support Agent Management

Business owners can delegate chats to support representatives. Under settings, tenants create sub-accounts by specifying a name, email, username, and password. Agents log in to their own dashboard to handle live chat handoffs and answer customer messages.

1.8 Chat Widget Customization & Embedding

To match the company's branding, the chat widget offers extensive customization options. Tenants can customize the widget title, header colors, background shades, message bubble colors, and upload a custom avatar. Once customized, tenants copy the auto-generated HTML script snippet and paste it right before the closing body tag of their website.

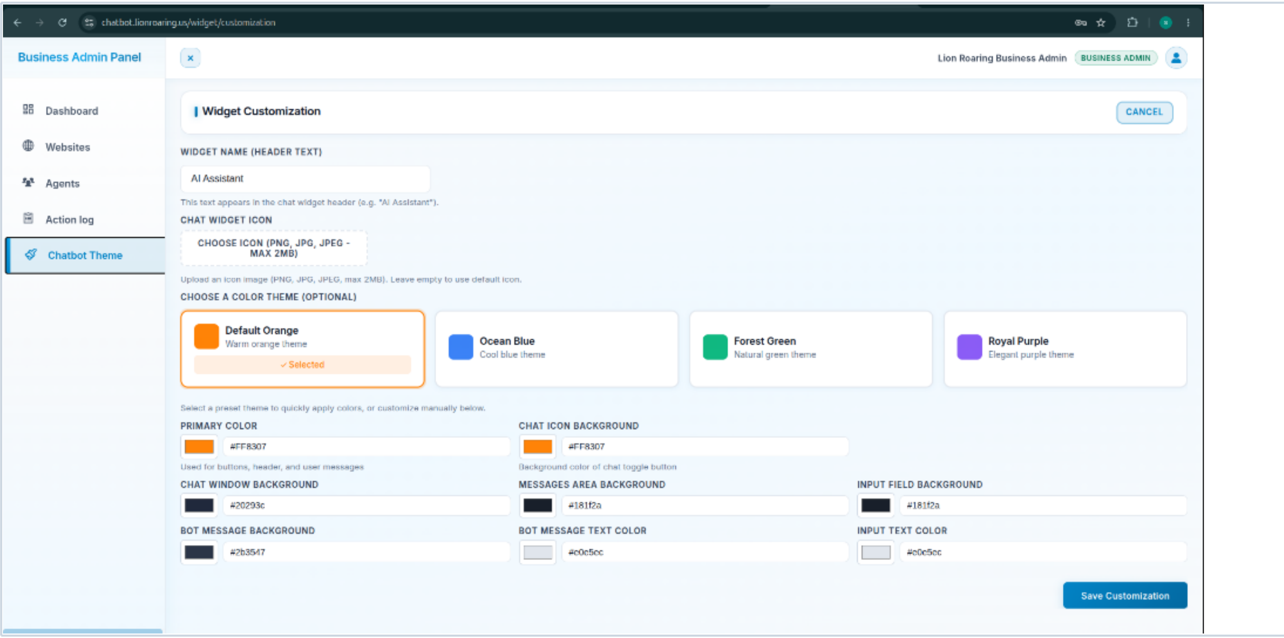

Figure 1.7: Widget Theme Customization Interface

1.9 Voice Chat Integration

Visitors can communicate with the chatbot using voice. The widget integrates browser-native Speech-to-Text, enabling users to speak into their microphone to type messages. Response synthesis (Text-to-Speech) reads the chatbot's answers aloud, with an option to switch between male and female voices.

1.10 Live Support Handoff & Messaging Queue

The chat widget operates in two modes. By default, the chatbot answers visitor questions. If the visitor requests a human agent, the widget triggers a handoff. If support agents are online and marked as available, the conversation is queued. When an agent accepts the chat, the status changes to human mode, and messages are sent back and forth in real-time via Socket.IO.

Agents can enable file uploads during a chat, allowing visitors to attach documents (up to 5MB). Sound notifications play on the agent console when new messages arrive.

Platform Limitations

> The platform does not support group chats, message reactions, mentions, or message modifications/deletions. Audio and video calling, screen sharing, call recordings, and live captions are not implemented.

2. Admin Guide

2.1 System Administration

Platform administrators log in to the same portal but have system-wide access. The Admin Dashboard displays consolidated platform metrics, including total registered business accounts, system-wide chatbots, active schedulers, database connection pool statistics, and audit logs.

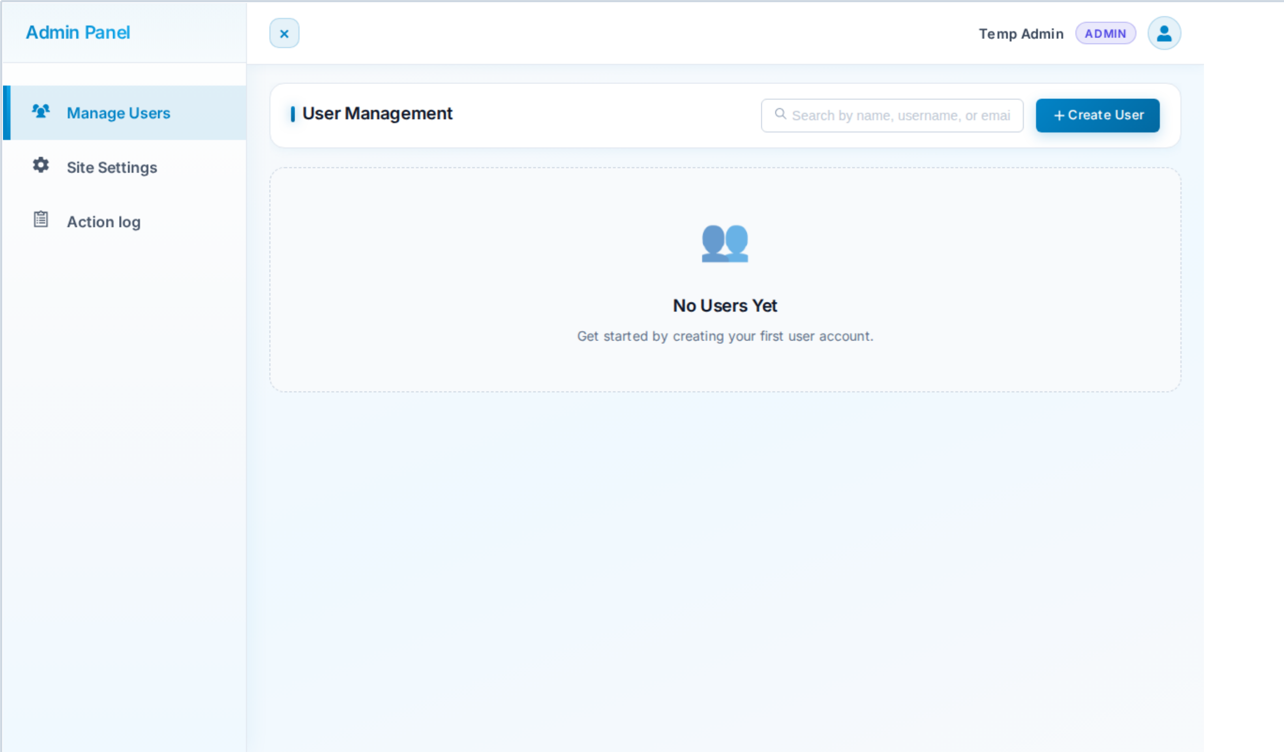

Figure 2.1: System User Administration Console

2.2 Tenant Provisioning & Limits

Administrators manage business tenants by adjusting settings such as bot and agent limits. The system enforces limits on the maximum bots a tenant can deploy (default 1, max 10) and the maximum agent accounts they can create (default 0, max 50). Suspending a tenant deactivates all logins immediately, while deleting a tenant removes all associated database records, scheduler processes, and vector folders.

2.3 System Audit & Action Logs

The platform logs administrative actions. Captured events include registrations, login/logout events, crawler status triggers, and agent updates. Logs are immutable and stored in the primary database, providing a reliable audit trail for compliance and support.

2.4 Maintenance & Configuration Settings

Administrators manage configurations using environment variables. These settings control database connection parameters, backend port bindings, token secrets, and fallback thresholds. When a tenant updates their knowledge base, administrators can trigger vector reloads to refresh the bot memory cache.

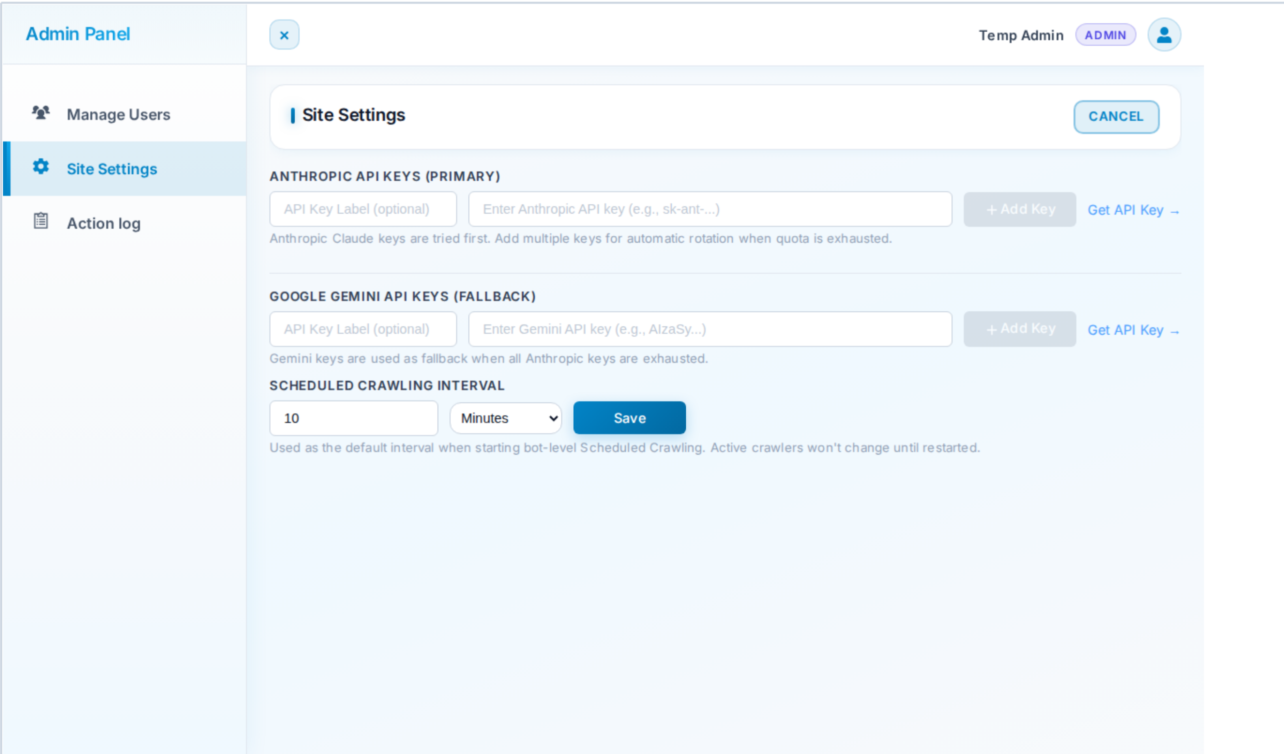

Figure 2.2: Site Configuration & API Keys Settings

3. AI Usage Guide

3.1 Retrieval-Augmented Generation (RAG)

The chatbot answers user queries by matching them against website content. The RAG pipeline processes queries in real-time:

1. The visitor sends a query.
2. The system vectorizes the query using the semantic embedding model.
3. A similarity search in the tenant's isolated database retrieves matching text passages.
4. A reranking model re-eval-u-ates and ranks the retrieved passages to find the most relevant context.
5. The top-ranked passages are combined with a system prompt and sent to the LLM (Anthropic Claude).
6. The LLM generates a response based on the provided context, complete with source citations.

3.2 Key Fallbacks & Performance

The system uses Anthropic Claude as the primary model and falls back to Google Gemini if Claude keys hit rate limits. Response latency averages between 1.2 and 3.0 seconds, depending on the models used. Reranking can be disabled to speed up response times.

3.3 Data Privacy

Tenant data is isolated. Vector files and databases are segregated, ensuring no data leaks between different businesses. The embedding models run locally, so raw content is not shared with external APIs for vectorization.

4. System Architecture

4.1 Architecture Stack

The system uses a containerized multi-tier architecture:

- **Frontend:** A React Single Page Application that handles user dashboards and widget rendering.
- **Backend:** An Express API server that manages authentication, Socket.IO messaging, and database routing.
- **RAG Bot Engine:** A FastAPI service that manages vector embedding generation and handles LLM completions.
- **Scraper Worker:** A Python service that manages background scraping tasks.

4.2 Multi-Tenant Resource Isolation

Database connections are pooled dynamically. The backend caches tenant connections in memory using connection strings. Connections are automatically closed after 60 seconds of inactivity to save resources. Vector files are saved in separate filesystem directories for each tenant, ensuring isolation.

5. Developer Guide

5.1 Environment Configuration

Developers configure the platform using a file containing keys for database connections, token secrets, and API credentials.

5.2 Local Installation

Setting up the development environment requires Node.js and Python. Developers install dependencies inside a Python virtual environment and run the backend, bot, worker, and frontend services locally on dedicated ports.

5.3 Docker Compose Deployments

The application runs in containers using Docker Compose. Building the stack compiles the React frontend and starts all services. In production, an external Nginx proxy routes traffic to the backend, frontend, and bot containers.

6. AI Prompt Guide

6.1 Synthesis Prompt

This prompt instructs the LLM to act as a customer support assistant. It restricts responses to the provided context, enforces a 50-word limit, and redirects visitors to human agents when requested.

6.2 Matcher & Translation Prompts

The matcher prompt maps visitor queries to the most relevant page URL. The translation prompt translates system messages into the visitor's language, caching results to improve response times.

7. FAQ & Troubleshooting

Q: Why is my website crawl status stuck on "Running"?

A: The crawler may be stuck in a redirection loop. Check the log file in the tenant's vector directory, and reduce the crawl depth setting.

Q: The bot is replying that it doesn't know the answer. How can I fix this?

A: The bot only answers questions based on indexed content. Verify that the website crawled successfully, or add manual Q&A items under the knowledge tab.

8. Glossary

- **RAG (Retrieval-Augmented Generation):** An AI workflow that retrieves relevant context from a database to generate accurate answers.
- **ChromaDB:** The vector database used to store and search page embeddings.
- **Handoff:** The process of transferring a visitor's active chat session from AI to a live agent.